

Project Title

Render Rush: 3D Endless Escape..

Core Game Idea

The player controls a character running down an endless road with three lanes. The road is filled with obstacles, coins, and magnets. The main goal is to survive as long as possible, get a high score, and avoid an enemy.

To do this, the player must:

- Move between the three lanes.
- Jump over or slide under different obstacles.
- Collect coins to get points.
- Collect magnets to easily pull items toward the player.
- Use a Hoverboard for special movements.
- Press specific keys in a row to multiply their score.
- Survive being chased by an enemy that appears when the player makes a mistake.

Why This Project Is Good

It includes:

- **Endless World:** The game automatically creates the road, obstacles, and coins as you run, so it never stops.
- **Realistic Body Movements:** The player and the enemy are built using connected body parts. When they run, jump, or slide, their arms and legs move naturally.
- **Smart Enemy:** The enemy does not just follow the player blindly. It looks at the road ahead and knows when to jump or slide to avoid its own obstacles.
- **Two Camera Views:** You can play by looking from behind the player (Third-Person) or by looking through the player's eyes (First-Person). In First-Person, the camera shakes like a real person running.
- **Combo System:** It rewards the player for pressing the right buttons quickly in a specific order.

Game Story

You are a futuristic runner on an endless, high-speed skyway. The road is full of dangerous barriers and scattered coins. If you make a mistake and hit a barrier, a robot guard is sent to chase you down. To stay alive, you must use your quick reflexes, a special hoverboard, and magnets to collect points and outrun the robot.

Main Gameplay Loop

The game flows step-by-step like this:

1. The player starts running in the middle lane.
2. The game slowly speeds up, making it harder.
3. Obstacles, coins, and magnets randomly appear in front of the player.
4. The player changes lanes, jumps, or slides to avoid crashing and to grab items.
5. If the player grabs a magnet, it pulls nearby coins toward them for a short time.
6. If the player hits an obstacle, they lose one life, and the enemy starts chasing them.
7. The enemy actively follows the player's lane and jumps/slides over its own obstacles.
8. The player can do a quick combo (Jump -> Switch Lane -> Slide) to earn 5 times the normal points.
9. The player can turn on the Hoverboard to change how they move.
10. The game continues until the player loses all 3 lives, which triggers "Game Over."

Assignment System Upgrade

Basic Game Feature	Upgraded Feature
Simple Movement	Player snaps exactly to 3 lanes, with realistic gravity when jumping and sliding.
Basic Score	Score increases with distance, plus extra points from coins and combo multipliers.
Basic Collision	The game accurately checks if the player is standing, sliding, or on a hoverboard before deciding if they hit an object.

Simple Game Over	The player has 3 lives. Losing a life spawns a smart enemy that chases the player and dodges its own obstacles.
Two Cameras	Players can switch between Third-Person and First-Person cameras.
Basic UI	The screen shows real-time updates for score, lives, camera mode, and combo alerts.
Simple Shapes	The characters have moving arms and legs, and the items (like coins and magnets) spin in 3D.